

EXPERIMENT – 35

Implementation of One-Time Pad Version of the Vigenère Cipher

CODE:

```
#include <stdio.h>

int main()
{
    char text[100], encrypted[100], decrypted[100];
    int key[100], n, i;

    printf("Enter plaintext: ");
    scanf("%s", text);

    printf("Enter key values: ");
    for (i = 0; text[i] != '\0'; i++)
        scanf("%d", &key[i]);

    n = i;

    /* Encryption */
    for (i = 0; i < n; i++)
        encrypted[i] = ((text[i] - 'A' + key[i]) % 26) + 'A';

    encrypted[n] = '\0';

    /* Decryption */
    for (i = 0; i < n; i++)
```

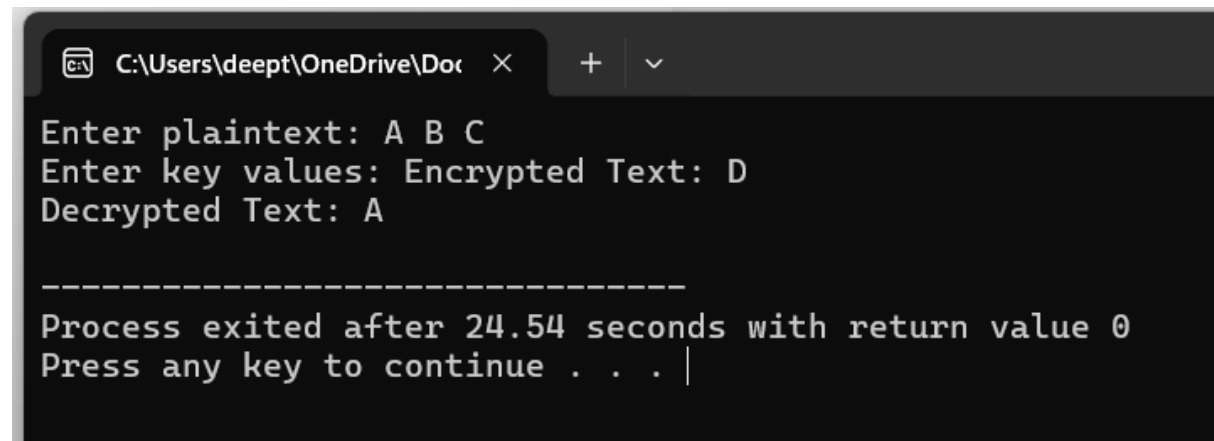
```
    decrypted[i] = ((encrypted[i] - 'A' - key[i] + 26) % 26) + 'A';

decrypted[n] = '\0';

printf("Encrypted Text: %s\n", encrypted);
printf("Decrypted Text: %s\n", decrypted);

return 0;
}
```

OUTPUT:



```
C:\Users\deept\OneDrive\Doc > Enter plaintext: A B C
Enter key values: Encrypted Text: D
Decrypted Text: A

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Process exited after 24.54 seconds with return value 0
Press any key to continue . . . |
```